PYTHON MINI PROJECT REPORT

TEMPERATURE CONVERTER

Student Name: _____
Roll Number: _____
Class / Division: _____
Subject: Python Programming
Submitted To: _____

1. Introduction

The Temperature Converter is a simple Python mini project. It converts temperature from Celsius to Fahrenheit and from Fahrenheit to Celsius.

2. Problem Statement

Temperature values may be given in different units. This project provides a simple way to convert a temperature from one unit to the other using Python.

3. Objectives

• To accept a temperature value from the user.
• To convert Celsius into Fahrenheit.
• To convert Fahrenheit into Celsius.
• To display the converted temperature.

4. Software Requirements

• Programming language: Python 3
• Editor: IDLE, VS Code, PyCharm or an online Python compiler
• Hardware: Computer or laptop

5. Methodology / Working

The program displays two conversion options. The user selects an option and enters a temperature. If option 1 is selected, the program applies the Celsius-to-Fahrenheit formula. If option 2 is selected, it applies the Fahrenheit-to-Celsius formula. The converted value is then displayed.

6. Conversion Formulas

Fahrenheit = (Celsius × 9 / 5) + 32

Celsius = (Fahrenheit – 32) × 5 / 9

7. Python Concepts Used

• Variables – store the entered and converted temperatures.
• input() – accept the user's choice and temperature.
• float() – convert input into a decimal number.
• if-elif-else – select the conversion method.
• Arithmetic operators – perform the conversion calculations.

• print() – display the result.

8. Sample Input and Output

Sample input:
Choice: 1
Temperature in Celsius: 25

Sample output:
Temperature in Fahrenheit: 77.0 F

9. Testing

Test 1: Convert 0°C. Expected result: 32°F.
Test 2: Convert 100°C. Expected result: 212°F.
Test 3: Convert 32°F. Expected result: 0°C.
Test 4: Enter an option other than 1 or 2. Expected result: Invalid choice message.

10. Conclusion

The Temperature Converter successfully converts temperatures between Celsius and Fahrenheit. This project helped me understand variables, user input, conditional statements and arithmetic operators in Python.

11. GitHub Repository

GitHub Repository Link: _____

Upload the Python source file and this report to your own GitHub repository, then paste the repository URL here.

APPENDIX – SOURCE CODE

```
# Temperature Converter
# This program converts Celsius to Fahrenheit and Fahrenheit to Celsius.

print("===== TEMPERATURE CONVERTER =====")
print("1. Celsius to Fahrenheit")
print("2. Fahrenheit to Celsius")

choice = input("Enter your choice (1 or 2): ")

if choice == "1":
    celsius = float(input("Enter temperature in Celsius: "))
    fahrenheit = (celsius * 9 / 5) + 32
    print("Temperature in Fahrenheit:", fahrenheit, "F")

elif choice == "2":
    fahrenheit = float(input("Enter temperature in Fahrenheit: "))
    celsius = (fahrenheit - 32) * 5 / 9
    print("Temperature in Celsius:", celsius, "C")

else:
    print("Invalid choice. Please select 1 or 2.")

print("Thank you for using Temperature Converter.")
```